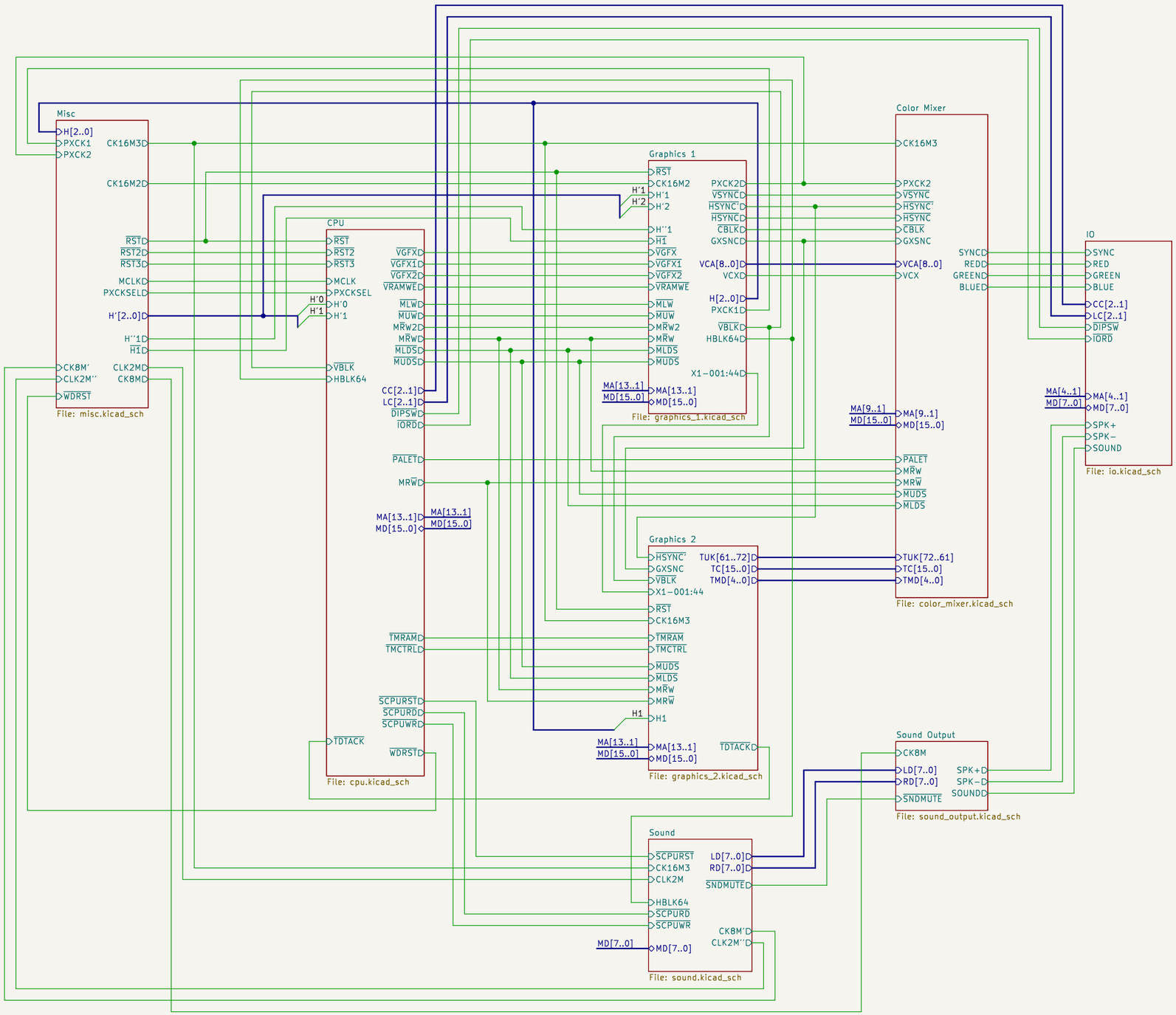


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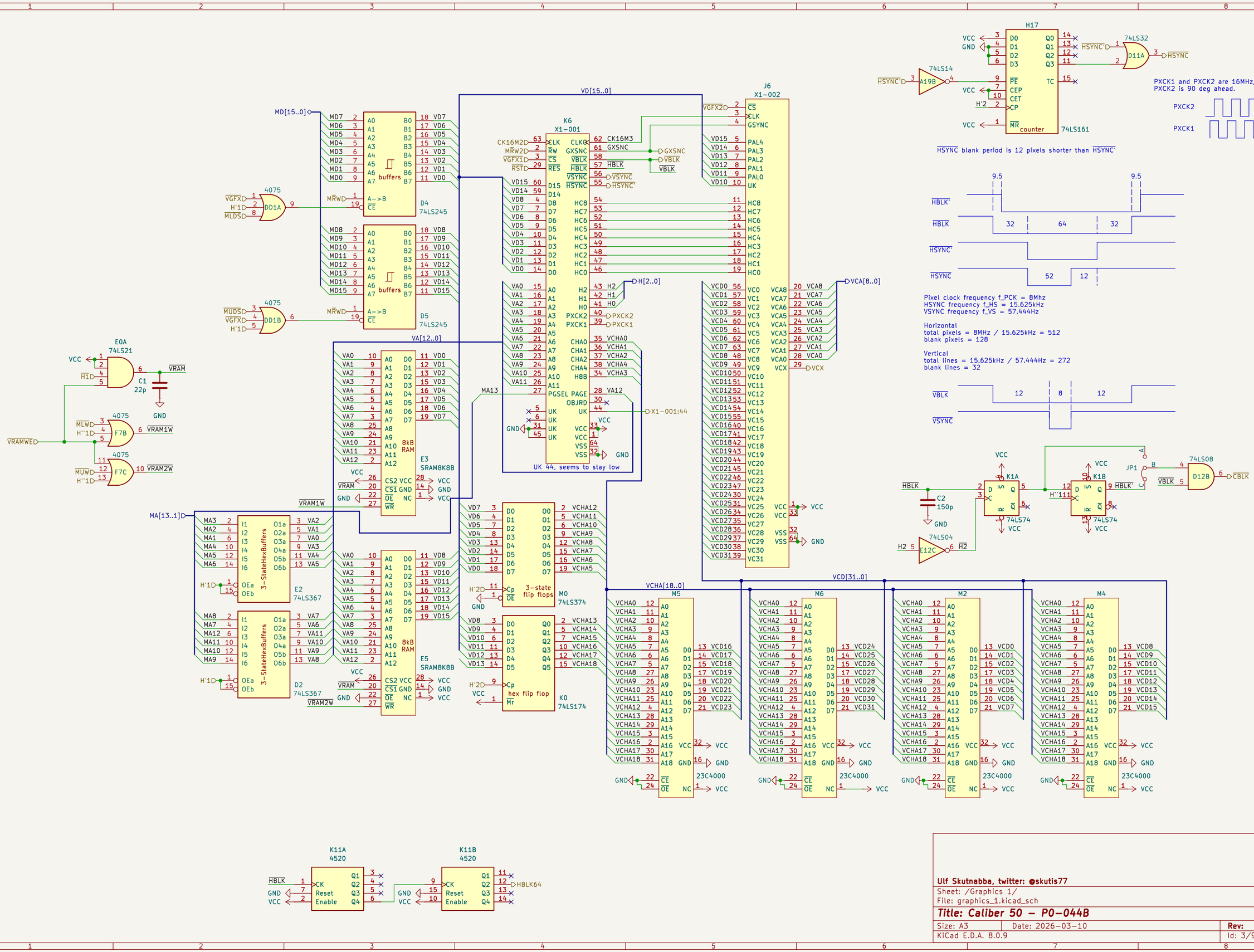
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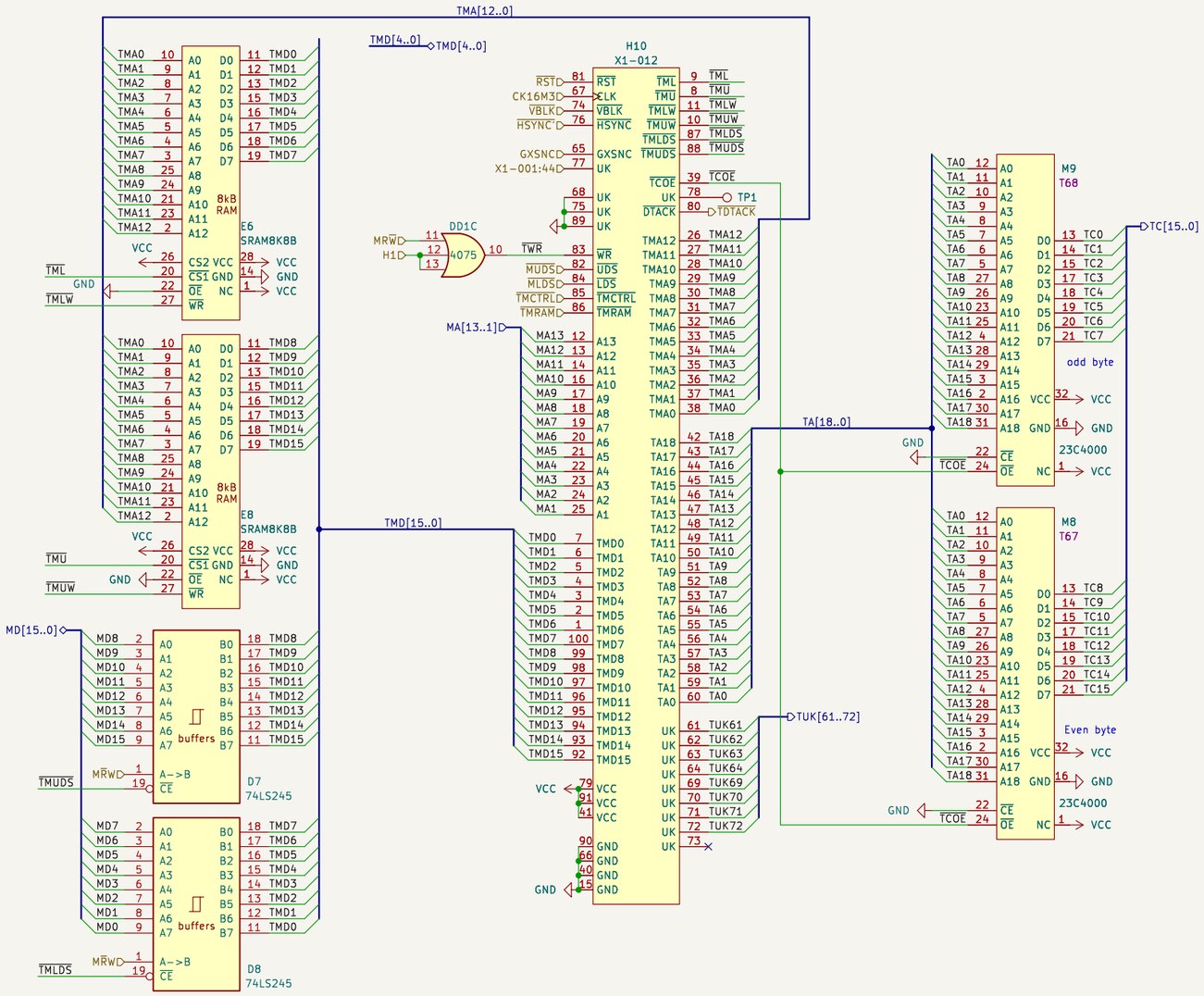
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Title: Caliber 50 – P0-044B

Size: A3	Date: 2026-03-10	**Rev:**
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B1
MC68000
MD[15..0]
MA[23..1]
MA[13..1]
MCLK
IPL2
IPL1
IPL0
BERR
HALT
RESET
DTACK
BGACK
BR
VPA
VMA
FC0
FC1
FC2
AS
UDS
LDS
RW
BG
MAS
MUDS
MLDS
MRW
RST2
RST
VCC
GND
C0A
C0B
74LS74
HIRST
VIRST
HBLK64
VBLKD
E12F
74LS04
VGFX
F4B
M1C
74LS00
H'1
H'0
PXCKSELD
TDTACK
E10A
74LS133
H2B
SNDRW
NVRAM
MRMDTCK
MRAM
PALET
DIPSW
COINCOIL
WDRST
VIRST
NVRAM
HIRST
E11C
74LS32
MFC
MAS
B6
B5
B11
B3
SRAM32K8B
23C4100
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
MA12
MA13
MA14
MA15
MRAM
MRW'
MLW
MUW
MROML
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
MD8
MD9
MD10
MD11
MD12
MD13
MD14
MD15
B12
SRAM2K8B
2kB
RAM
74LS00
CC12C
CC12B
E12D
E12E
NVRAM
MLW
RST3
VCCB
PWR_FLAG
D1
R7
C27
100u
BAT1
Battery_Cell
D10
74LS138
decoder
MA19
MA20
MA21
MA22
MA23
MROML
MROMH
E9
MRMDTCK
HIRST
NVRAM
VIRST
WDRST
COINCOIL
DIPSW
PALET
D9
TMCTRL
TMRAM
IORD
VGFX2
VRAMWE
MRAM
VGFX1
D0B
74LS139
E0B
74LS21
D11D
MLDS
F5C
74LS10
VGFX1'
VGFX2
VRAMWE
VGFX
H2A
F5B
MFC2
MFC1
MFC0
H2D
MFC is active during interrupts.
E11B
E11A
E11D
M1A
D12D
74LS08
74LS32
MRW2
MUW
SCPURD
SCPUWR
K13A
E13C
D6
74LS174
hex flip flop
CC2
LC2
CC1
LC1
SCPURST
CC[2..1]
LC[2..1]
JP5
JP4
B7
B9
27C512
Ulf Skutnabba, twitter: @skutis77
Sheet: /CPU/
File: cpu.kicad_sch
Title: Caliber 50 – P0-044B
Size: A3
Date: 2026-03-10
KiCad E.D.A. 8.0.9
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VD[15..0]
MD[15..0]
VA[12..0]
VA[18..0]
VCD[31..0]
MA[13..1]
VRAM1W
VRAM2W
VRAMWE
VCA[8..0]
H[2..0]
MA13
VCC
GND
VRAM
H1
H'1
H''1
H'2
H2
HBLK
VBLK
HSYNC
HSYNC'
VSYNC
CBLK
HBLK64
MLW
MUW
MUDS
MLDS
VGFX
VGFX2
MRWD
CK16M2
MRW2
VGFX1
RST
GXSNC
CK16M3
PXCK1
PXCK2
MA
VA
VD
VCD
VCA
VCHA
HC
PAL
UK
VC
H17
74LS161
counter
74LS32
D11A
74LS14
A19B
PXCK1 and PXCK2 are 16MHz/2 = 8MHz.
PXCK2 is 90 deg ahead.
HSYNC blank period is 12 pixels shorter than HSYNC'
9.5
32
64
52
12
Pixel clock frequency f_PCK = 8Mhz
HSYNC frequency f_HS = 15.625kHz
VSYNC frequency f_VS = 57.444Hz
Horizontal
total pixels = 8MHz / 15.625kHz = 512
blank pixels = 128
Vertical
total lines = 15.625kHz / 57.444Hz = 272
blank lines = 32
K6
X1-001
UK 44, seems to stay low
J6
X1-002
E0A
74LS21
C1
22p
F7B
F7C
E3
SRAM8K8B
E2
74LS367
D2
D4
74LS245
D5
E5
M0
74LS374
K0
74LS174
M5
M6
M2
M4
23C4000
K11A
4520
K11B
C2
150p
74LS04
E12C
K1A
K1B
74LS74
JP1
74LS08
D12B
Ulf Skutnabba, twitter: @skutis77
Sheet: /Graphics 1/
File: graphics_1.kicad_sch
Title: Caliber 50 - P0-044B
Size: A3
Date: 2026-03-10
Rev:
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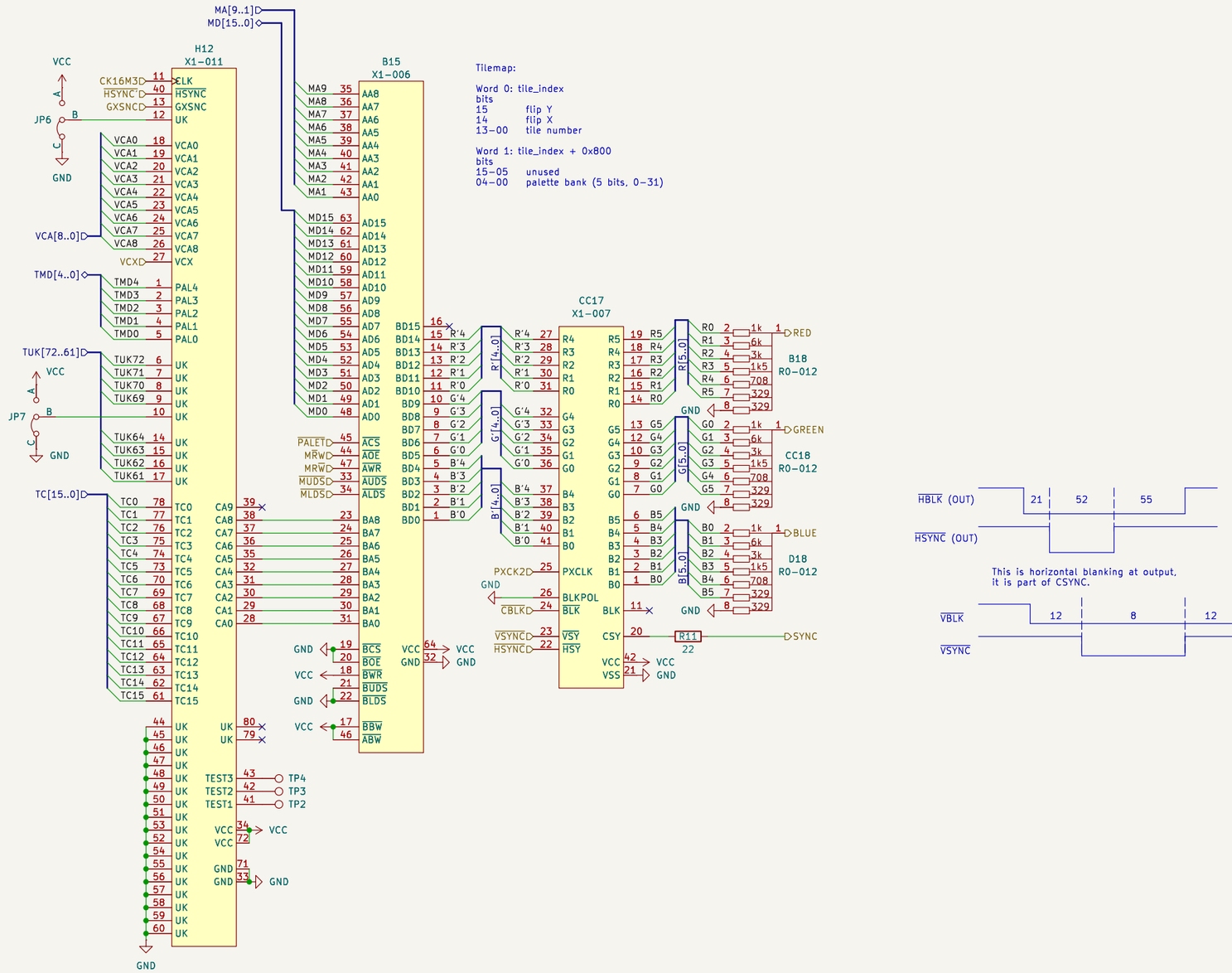


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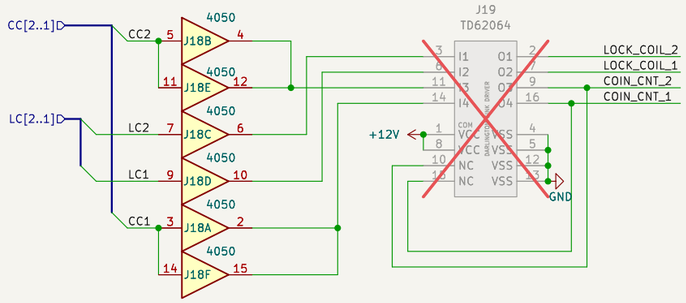
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File: graphics_2.kicad_sch

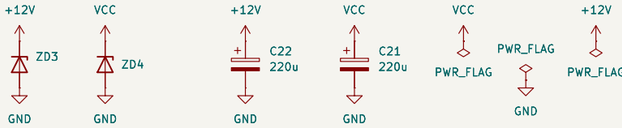
Title: Caliber 50 – P0-044B

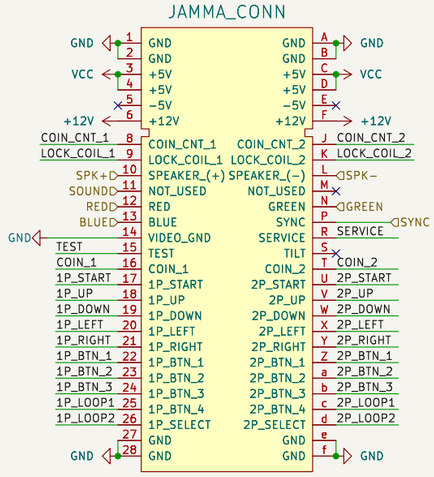
Size: A3	Date: 2026-03-10	**Rev:**
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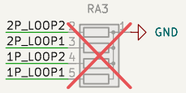


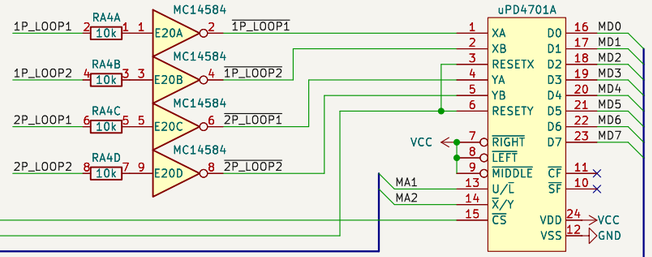
Ulf Skutnabba, twitter: @skutis77	
Sheet: /Color Mixer/	
File: color_mixer.kicad_sch	
Title: Caliber 50 – P0-044B	
Size: A3	Date: 2026-03-10
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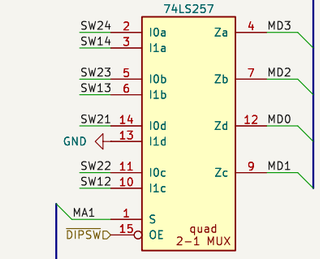


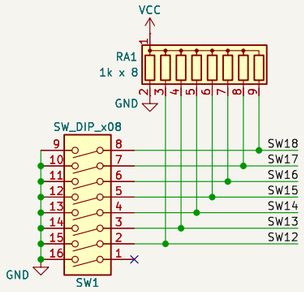


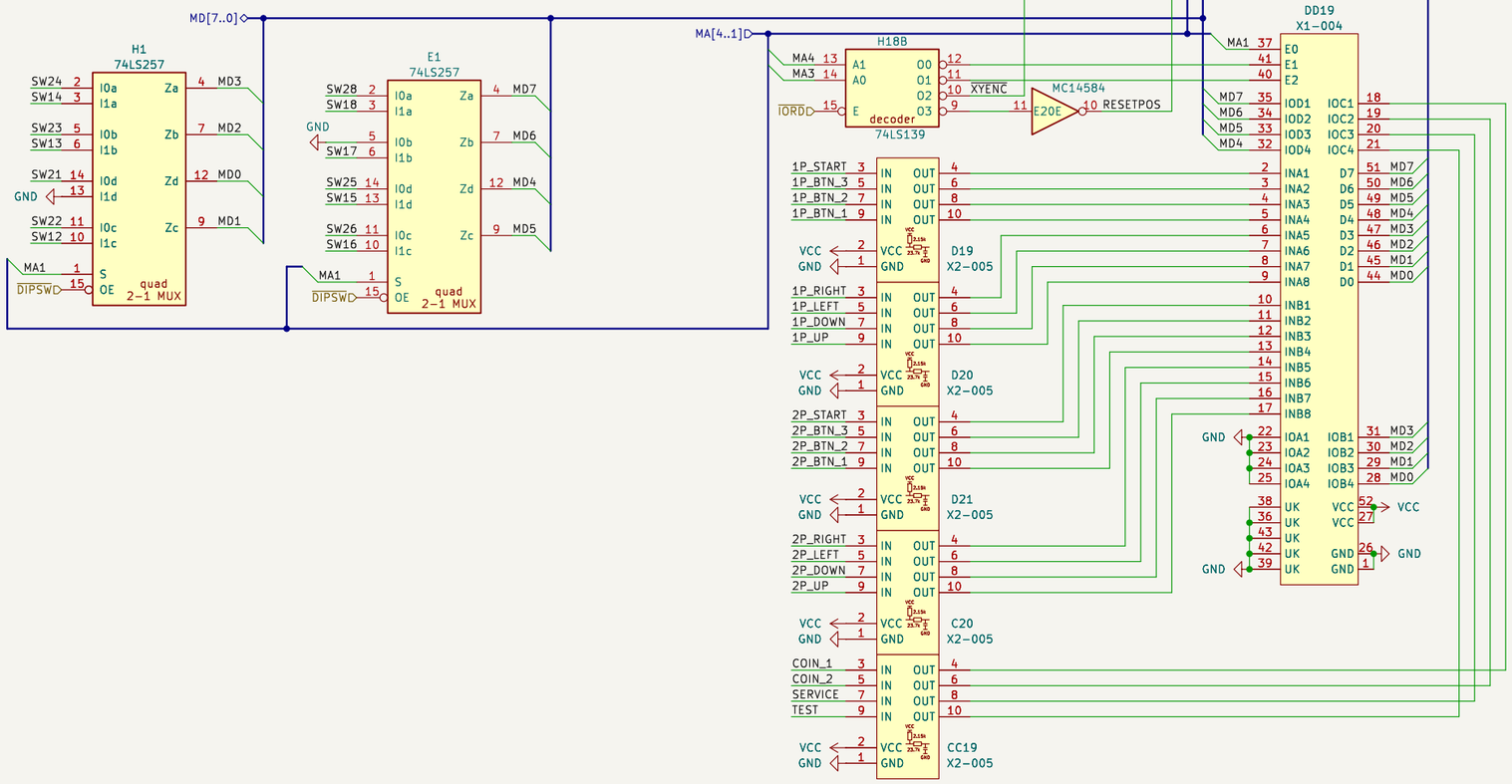












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Sheet: /IO/

File: io.kicad_sch

Title: Caliber 50 – P0-044B

Size: A3	Date: 2026-03-10	Rev:
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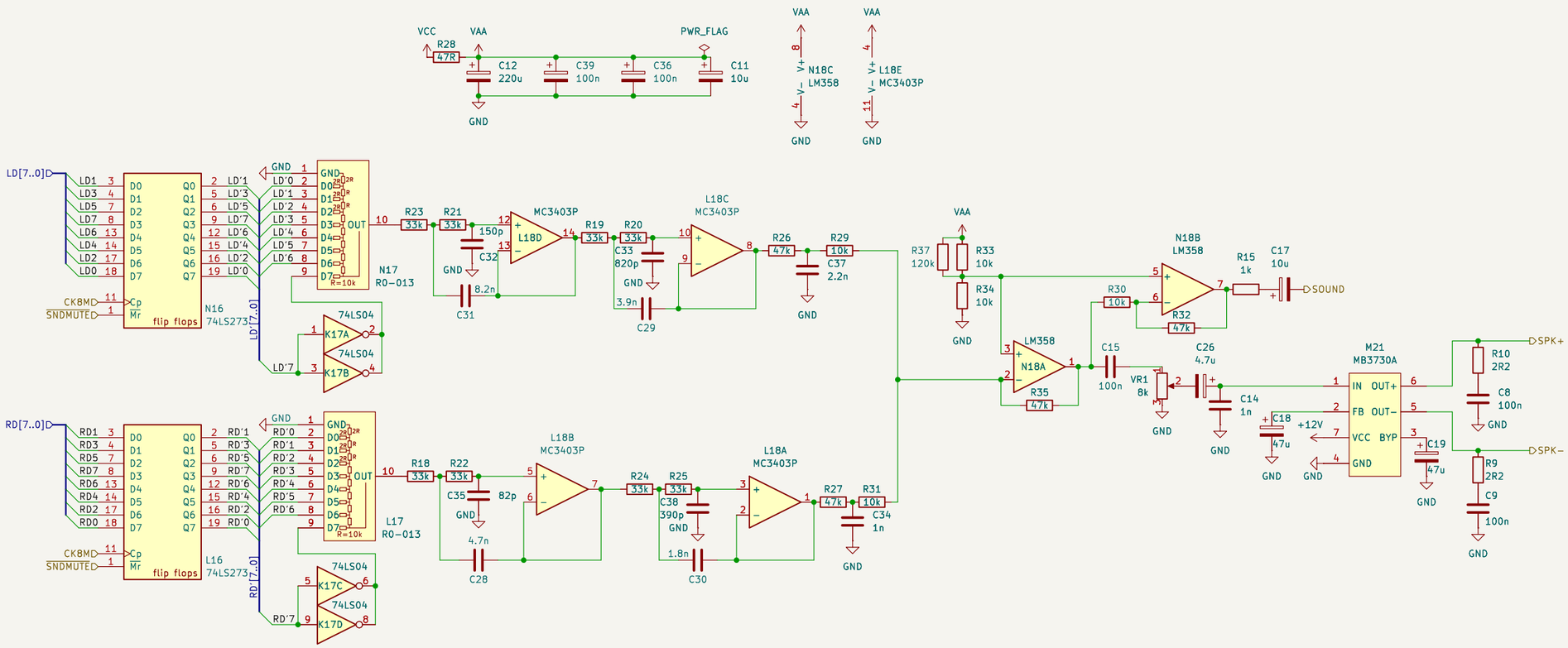
MD[7..0]
MD7 3
MD6 4
MD5 7
MD4 8
MD3 13
MD2 14
MD1 17
MD0 18
D0 D1 D2 D3 D4 D5 D6 D7
O0 O1 O2 O3 O4 O5 O6 O7
2 SD7
5 SD6
6 SD5
9 SD4
12 SD3
15 SD2
16 SD1
19 SD0
3-state flip flops
D15
74LS374
SRD 1
E13A
2
74LS32
SCPUWR 11
3
Cp
OE
1
SD7 3
SD6 4
SD5 7
SD4 8
SD3 13
SD2 14
SD1 17
SD0 18
D0 D1 D2 D3 D4 D5 D6 D7
O0 O1 O2 O3 O4 O5 O6 O7
2 MD7
5 MD6
6 MD5
9 MD4
12 MD3
15 MD2
16 MD1
19 MD0
MD[7..0]
3-state flip flops
E15
74LS374
SCPURD 1
Cp
OE
11
74LS32
K13B
4
5
6
SWR
H18A
SA15 3
SA14 2
A1
A0
E
1
O0 O1 O2 O3
4 PCMRAM
5
6
7
decoder
74LS139
GND
SD7 3
SD6 4
SD5 7
SD4 8
SD3 13
SD2 14
SD1 17
SD0 18
D0 D1 D2 D3 D4 D5 D6 D7
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
2 BNK3
5 BNK2
6 BNK1
9 BNK0
12 SNMICLR
15 SIRQCLR
16 SNDMUTE
19
flip flops
F14
74LS273
74LS32
E13D
12
13
11
SWR
SCPURST 1
Cp
Mr
BNK[3..0]
BNK3 2
BNK2 5
BNK1 11
BNK0 14
Quad Mux
I0a I0b I0c I0d
I1a I1b I1c I1d
SEL
G
Za Zb Zc Zd
4 RA17
7 RA16
9 RA15
12 RA14
3
6
10
13
GND
1
15
GND
K14
74LS157
SA[15..0]
PCMD7 2
PCMD6 3
PCMD5 4
PCMD4 5
PCMD3 6
PCMD2 7
PCMD1 8
PCMD0 9
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
18 SD7
17 SD6
16 SD5
15 SD4
14 SD3
13 SD2
12 SD1
11 SD0
buffers
A->B
CE
L15
74LS245
PCMBUSZ 9
PCMRAM 10
74LS32
K13C
8
SRW 1
19
VCC 4
SA12 5
74LS00
K12B
6
SA11 3
SA10 4
SA9 5
SA8 6
SA7 7
SA6 8
SA5 9
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
18 PCMA12
17 PCMA11
16 PCMA10
15 PCMA9
14 PCMA8
13 PCMA7
12 PCMA6
11 PCMA5
buffers
A->B
CE
VCC 1
PCMBUSZ 19
L14
74LS245
PCMA[19..0]
SA4 2
SA3 4
SA2 6
SA1 10
SA0 12
I1 I2 I3 I4 I5 I6
O1a O2a O3a O4a O5b O6b
3 PCMA4
5 PCMA3
7 PCMA2
9 PCMA1
11 PCMA0
13
3-StateHexBuffers
GND 14
OEa OEb
1
15
PCMBUSZ
N14
74LS367
PCMD[7..0]
X1-010
CK16M3 1
CLK
SRW 6
PCMRAM 5
SCPUWR 3
RW
RAM
BUSINH
CLKD2 2
CLKD8 80
CK8M'
CLK2M''
BUSZ 8 PCMBUSZ
RAMOE 43 PCMRAMOE
RAMWR 42 PCMRAMWR
RAMEN 41 PCMRAMEN
ROMEN 40 PCMROMEN
PCMD7 78
PCMD6 77
PCMD5 76
PCMD4 75
PCMD3 74
PCMD2 71
PCMD1 70
PCMD0 69
D7 D6 D5 D4 D3 D2 D1 D0
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16 A17 A18 A19
46 PCMA0
47 PCMA1
48 PCMA2
49 PCMA3
50 PCMA4
53 PCMA5
54 PCMA6
55 PCMA7
56 PCMA8
57 PCMA9
58 PCMA10
59 PCMA11
60 PCMA12
61 PCMA13
63 PCMA14
64 PCMA15
65 PCMA16
66 PCMA17
67 PCMA18
68 PCMA19
GND 39 38 37 36 35 34
VCC 23
GND 7
VCC 72 52 33 13
GND 73 62 51 32 12
UK 44 45 22 11
UK 4 9 10 79
LD0 14 LD1 15 LD2 16 LD3 17 LD4 18 LD5 19 LD6 20 LD7 21
LD[7..0]
RD0 24 RD1 25 RD2 26 RD3 27 RD4 28 RD5 29 RD6 30 RD7 31
RD[7..0]
J15
PCMA0 12
PCMA1 11
PCMA2 10
PCMA3 9
PCMA4 8
PCMA5 7
PCMA6 6
PCMA7 5
PCMA8 27
PCMA9 26
PCMA10 23
PCMA11 25
PCMA12 4
PCMA13 28
PCMA14 29
PCMA15 3
PCMA16 2
PCMA17 30
PCMA18 31
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16 A17 A18
D0 D1 D2 D3 D4 D5 D6 D7
13 PCMD0
14 PCMD1
15 PCMD2
17 PCMD3
18 PCMD4
19 PCMD5
20 PCMD6
21 PCMD7
M12
T68
32 VCC
16 GND
CE
OE
PCMROMEN 22
PCMA19 24
NC
1 VCC
23C4000
M11
T68
PCMA19 1
E12A
2
74LS04
PCMA0 10
PCMA1 9
PCMA2 8
PCMA3 7
PCMA4 6
PCMA5 5
PCMA6 4
PCMA7 3
PCMA8 25
PCMA9 24
PCMA10 21
PCMA11 23
PCMA12 2
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12
D0 D1 D2 D3 D4 D5 D6 D7
11 PCMD0
12 PCMD1
13 PCMD2
15 PCMD3
16 PCMD4
17 PCMD5
18 PCMD6
19 PCMD7
8kB RAM
M13
SRAM8K8B
VCC
PCMRAMEN 20
PCMRAMOE 22
PCMRAMWR 27
CS2 VCC
CS1 GND
OE NC
WR
28 VCC
14 GND
1 VCC
SD[7..0]
E16
R65C02
SD0 33
SD1 32
SD2 31
SD3 30
SD4 29
SD5 28
SD6 27
SD7 26
D0 D1 D2 D3 D4 D5 D6 D7
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
9 SA0
10 SA1
11 SA2
12 SA3
13 SA4
14 SA5
15 SA6
16 SA7
17 SA8
18 SA9
19 SA10
20 SA11
22 SA12
23 SA13
24 SA14
25 SA15
SCPURSTD 40
RES
IRQ 4
NMI 6
NC 5 35 36
RDY SYNC 7
CLK2MD 37
φ0 φ1 φ2
3
39 CLK2M'
VCC 8
VSS 1 21
R/W 34 SRW
SO 38
VCC 2
GND
SIRQCLR
J14B
GND 12
HBLK64 11
D
C
Q
9
8
74LS74
VCC
SNMICLR
J14A
GND 2
SCPUWR 3
5
6
74LS74
VCC
RA[17..14]
SA0 12
SA1 11
SA2 10
SA3 9
SA4 8
SA5 7
SA6 6
SA7 5
SA8 27
SA9 26
SA10 23
SA11 25
SA12 4
SA13 28
RA14 29
RA15 3
RA16 2
RA17 30
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16 A17 A18
D0 D1 D2 D3 D4 D5 D6 D7
13 SD0
14 SD1
15 SD2
17 SD3
18 SD4
19 SD5
20 SD6
21 SD7
E17
005
32 VCC
16 GND
CE
OE
SRD 24
NC
1 VCC
23C4000
SA15 3
E12B
4
74LS04
74LS00
CLK2M' 9
SRW 10
K12C
8 SRD
74LS00
CLK2M' 12
K12D
11 SWR
74LS00
VCC 1
SRW 2
K12A
3 SRW 13

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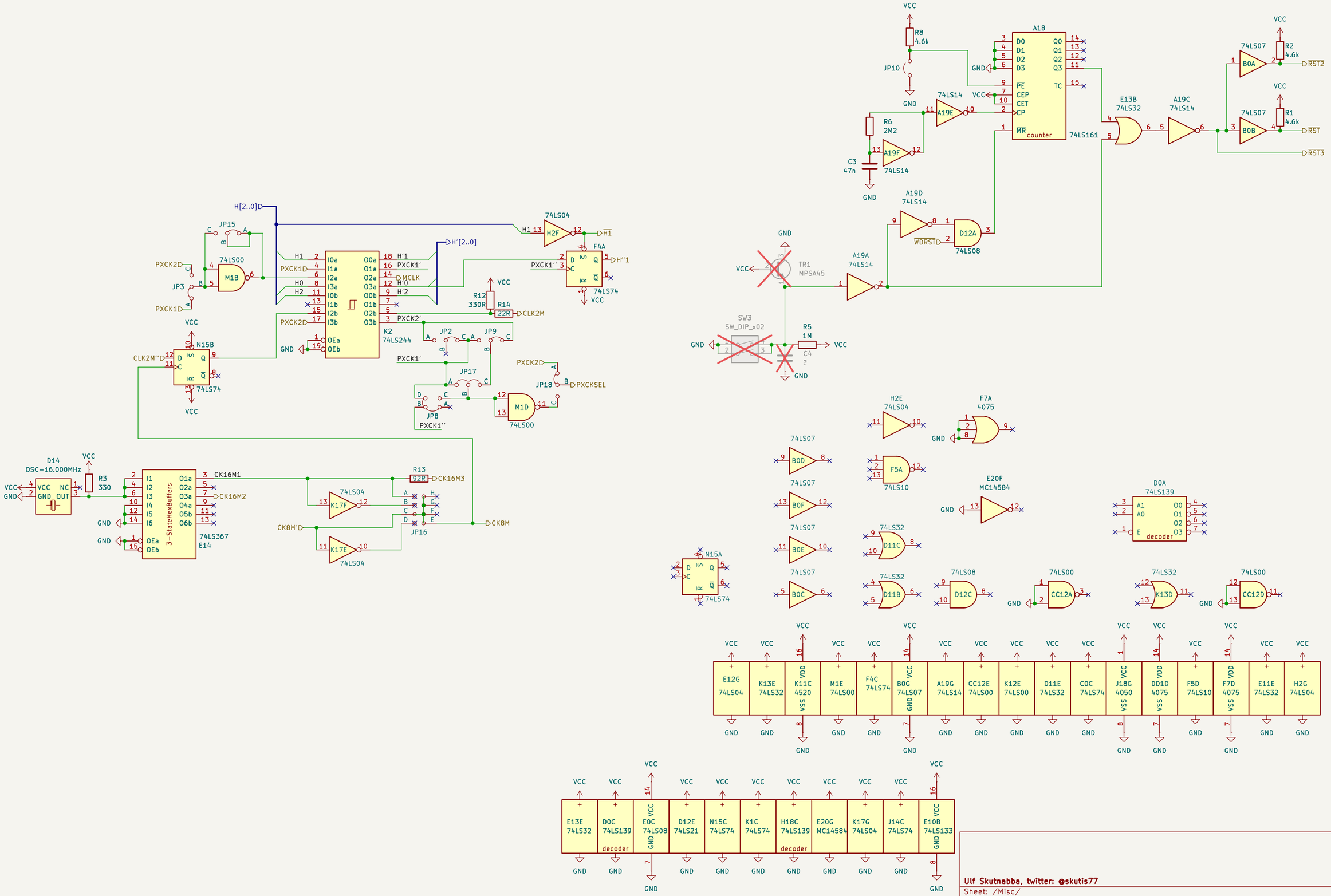
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Title: Caliber 50 – P0-044B

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Sheet: /Sound Output/	
File: sound_output.kicad_sch	
Title: Caliber 50 – P0-044B	
Size: A3	Date: 2026-03-10
KiCad E.D.A. 8.0.9	Id: 8/9
	Rev:



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Sheet: /Misc/

File: misc.kicad_sch

Title: Caliber 50 – P0-044B

Size: A3	Date: 2026-03-10	Rev:
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